

Emmanuel Gnabeyeu Mbiada

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Profile

Doctoral Researcher in Mathematics with applications to Mathematical finance and Artificial Intelligence at Sorbonne Université and Université Paris Cité (LPSM), passionate about bridging theory and applications. Research interests include Reinforcement Learning, Stochastic Analysis, Stochastic Optimal Control, Stochastic Optimization for Machine Learning, Bayesian Statistics, Langevin Score's based Generative Diffusion Models, Probabilistic Modelling and Probabilistic Machine Learning.

Academic Background

- 2024–2027 **Ph.D. Researcher in Mathematics and Applications** — Sorbonne Université & Université Paris Cité, Laboratoire de Probabilités, Statistique et Modélisation (LPSM), Paris, France.
- **Thesis:** *Stochastic Volterra Equations: Asymptotic Stationarity and Applications.*
- 2023–2024 **MPhil/M.Sc. in Applied Mathematics and Statistics (Track Probability and Finance)** — Sorbonne Université & Institut Polytechnique de Paris, France. *Highest Honours.*
- *Machine Learning and statistical models for multiscale market finance.*
 - **Thesis:** *Reinforcement Learning for Volatility Fitting.*
- 2020–2024 **Ingénieur Polytechnicien/MSc at École Polytechnique (I'X)**, one of France's top-ranking schools of science and engineering, Palaiseau, France.
- *Applied Mathematics, Computer Science and Economics.*
- Cumulated GPA: 3.8/4.0.**
- 2021–2022 **Bachelor's Degree in Philosophy** — Université Paris Nanterre, France.
- *Politics, Philosophy and Theology.* *High Honours.*
- 2017–2020 **B.Sc. in Mathematics** — Université de Yaoundé I, Cameroon.
- *High Honours; Applied and Computational Mathematics.*
- 2017–2020 **Preparatory Classes (Mathematics & Physics)** — National Advanced School of Engineering, Yaounde (ENSPY), Cameroon.
- *Mathematics and Physical Science.* *Highest Honours.*

Selected Publications and Preprints

- Gnabeyeu, E. & Souilmi, S.** Smile Dynamics under the fake stationary quadratic rough Heston Volatility: SPX and Skew-Stickiness Ratio, *in preparation*.
- September 2026 **Gnabeyeu, E. & Dro, S. B.** On explicit solutions to a class of quadratic BSDEJs driven by affine Volterra processes with jumps and applications, *working paper*.
- August 2026 **Gnabeyeu, E. & Pagès, G.** On (fake) Stationarity in Stochastic Volterra Equations with Affine Drift and Regular Kernels, *Submitted for publication*.
- May 2026 **Gnabeyeu, E. & Dro, S. B.** Optimal Merton's Problem under Multivariate Affine Volterra Models with Jumps, *Submitted for publication*
- April 2026 **Gnabeyeu, E. & Pagès, G.** On Path-dependent Volterra Integral Equations: Strong Well-posedness and Stochastic Numerics, *Submitted for publication*.

- March 2026 **Gnabeyeu, E.** On Utility Maximization under Multivariate Fake Stationary Affine Volterra Models, *Submitted for publication.*
- February 2026 **Gnabeyeu, E. & Pagès, G. & Rosenbaum, M.** Fake stationary rough Heston volatility: Microstructure-inspired foundations, *Submitted for publication.*
- November 2025 **Gnabeyeu, E. & Pagès, G. & Rosenbaum, M.** On Inhomogeneous Affine Volterra Processes: Stationarity and Applications to the Volterra Heston Model, *Submitted for publication.*
- September 2025 **Gnabeyeu, E. & Pagès, G.** On a Stationarity Theory for Stochastic Volterra Integral Equations with Affine Drift, *Submitted for publication.*
- October 2024 **Gnabeyeu, E., Karkar, O., Idboufous, I.** Solving the Dynamic Volatility Fitting Problem: A Deep Reinforcement Learning Approach, With Citigroup Global Markets Limited.

Conferences and Summer Schools

- June 29th – July 3rd, 2026 **XIII Bachelier World Congress of the Bachelier Finance Society 2026** — University of Bologna, Bologna, Italy.
- *Attended plenary talks by a distinguished lineup of international academics, practitioners, and researchers, covering recent advances and key emerging topics in financial mathematics.*
- 27–30 January 2026 **Conference on Advances in Financial Mathematics 2026 : directions and some information** — Sorbonne University, Campus Pierre et Marie Curie, Paris, France.
- *Attended plenary talks on recent advances in financial Mathematics, from both academic researchers and practitioners.*
- June–July 2025 **International Joint Conference on Neural Networks (IJCNN 2025)** — Pontifical Gregorian University, Rome, Italy.
- *Presenting the paper "Solving the Dynamic Volatility Fitting Problem: A Deep Reinforcement Learning Approach in Continuous State Action Spaces."*
 - *Attended workshops on Reinforcement Learning, Diffusion Models and Flow, Deep Generative Models, and Bayesian Methods for Inference and Learning.*
- July 2025 **5th Barcelona Summer School of Stochastic Analysis and Quantitative Finance** — Centre de Recerca Matemàtica (CRM), Autonomous University of Barcelona, Spain.
- *Participated in lectures on rough volatility and rough path theory, and their applications to signatures, machine learning, and stochastic optimal control.*

Selected Talks

- June 30th, 2026 **13th Bachelier World Congress of the Bachelier Finance Society** — University of Bologna - Bologna, Italy.
- *On the fake stationary rough volatility models.*
- June 2026 **Bachelier Seminar** — Institut Henri Poincaré - Paris, France.
- *A Journey Through Exponential Utility Maximization under Multivariate Fake Stationary Affine Volterra Models.*
- December 2025 **Doctoral Seminar** — Sorbonne Université, LPSM, Paris, France.
- *On Inhomogeneous Fractional Square-Root Process: Microstructural Foundation and Weak Stationarity Theory.*
- October 2025 **Natixis Foundation for Research and Innovation Awards Ceremony for Best Master's Thesis in Quantitative Finance** — Sky Garden (BPCE Tower), Paris, France.
- *A Reinforcement Learning Perspective to Exotics and Flow Equity Derivatives.*

July 2025 **5th Barcelona Summer School of Stochastic Analysis and Quantitative Finance** — Centre de Recerca Matemàtica, Autonomous University of Barcelona, Spain.

- *On a Stationarity Theory for Stochastic Volterra Integral Equation with Affine Drift: Finite-Time, Functional Weak Asymptotics and Applications.*

Teaching Experience

2025–May. 2026 **Teaching Assistant** — Sorbonne Université, Paris, France.

- *Course: Numerical Analysis (Bachelor's Level 3, Mathematics).*
- *Course: Advanced Mathematics (Bachelor's Degree – First Year, Mathematics).*

Sept.–Dec. 2025 **Teaching Assistant** — Université Paris 1 Panthéon-Sorbonne, Paris, France.

- *Course: Probabilities and Statistics (Bachelor Level 3, Economics).*

Sept.–Dec. 2025 **Teaching Assistant** — ESILV – Graduate Engineering School, Paris, France.

- *Course: Advanced Statistics (Master 1).*
- *Covered topics: Inferential and Bayesian Statistics, Non-parametric Statistic and Hypothesis Testing.*

2018–2020 **Teacher and Tutor** — Intelligencia Corporation.

- *Tutoring in mathematics and physics.*

Professional Experience

Apr.–Nov. 2024 **Quantitative Researcher** — Citigroup Global Markets Limited, London, UK.

- *Machine Learning for flows and exotic equity derivatives.*

Apr.–Sept. 2023 **Quantitative Researcher & Strategist** — BNP Paribas Asset Management, Paris, France.

- *Machine Learning for mid-frequency systematic equity investment strategies.*

Jun.–Sept. 2022 **Data Scientist & ML Engineer** — BOXY Start-Up, Paris, France.

- *Mathematical modelling and demand forecasting in autonomous retail systems.*

2021–2022 **Data Scientist Intern** — Alvidence Start-Up & École Polytechnique, Palaiseau, France.

- *Explainability of AI and statistical learning algorithms (XAI project).*

Awards, Honors, Distinctions and Scholarships

2025 **First Prize – Best European Master's Thesis in Quantitative Finance** — Natixis Foundation for Research and Innovation.

- *First Prize for the 2025 edition of the best master's thesis in Mathematical Finance.*

2021 – 2023 **Eiffel Excellence Scholarship.** — from the French Government.

2017 **Diploma of Excellence – National Scientific Olympiad** — Cameroon.

2011 **Africa Challenges Group Merit Scholarship.** — from the United States.

Skills

- **Computer Science:** Advanced in Python (Numpy, Pandas, Scikit-Learn, PyTorch), Intermediate in C++, SQL, knowledge in Java, MATLAB.
- **Languages:** French (native speaker), English (C1), German (B1), Spanish (A2).

Interests

- Reading (literature, philosophy, politics, theology).
- Travel, music, cinema, theatre, drawing, boxing.